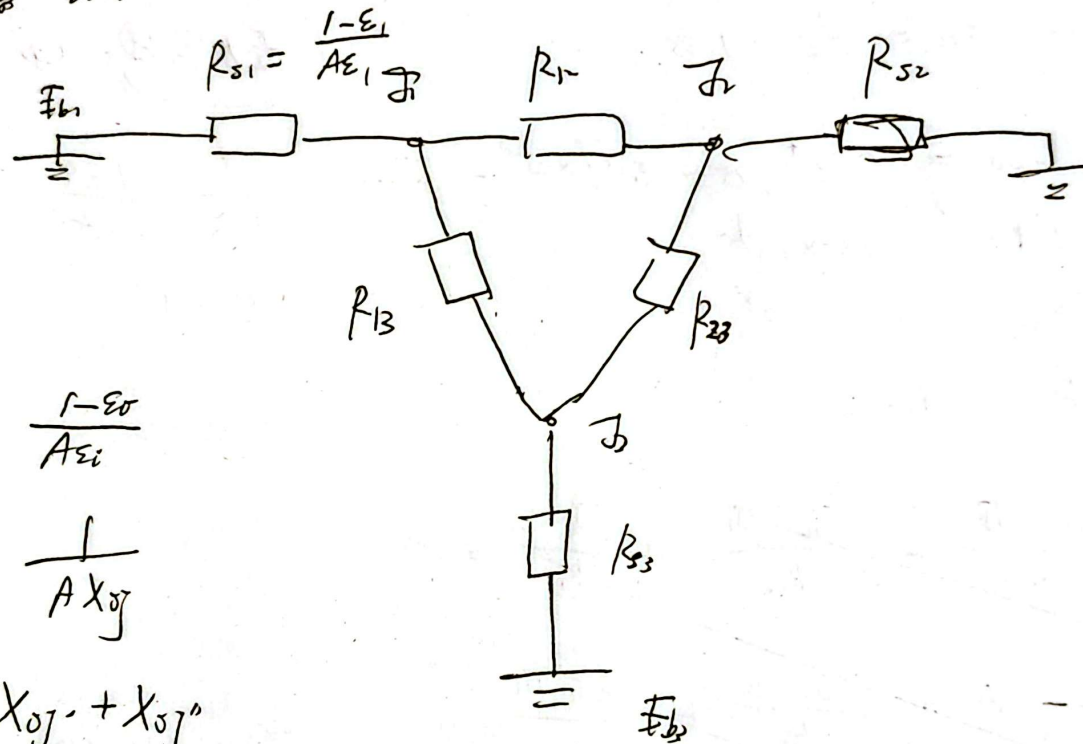


潘建浩 2023080041

S11-13



$$R_{s1} = \frac{1-\epsilon_1}{A\epsilon_1}$$

$$R_{ij} = \frac{1}{A X_{ij}}$$

$$X_{ij} = X_{ij}^* + X_{ij}^0$$

↓
查表

$J_3 = E_3$ (当且仅当 $R_{s3} = 0$, $\epsilon_3 = 1$, β 为黑体)

表面阻抗 (ϵ 灰体 < 1), 反映 “非集非黑体特性”

$$R_{s1} = \frac{1-\epsilon_1}{A\epsilon_1} \quad \text{对辐射耦合电阻}$$

空间辐射阻抗

$$R_{space} = \frac{1}{A_0 X_{ij}} \quad \text{反映 “空间几何位置”}$$

对辐射耦合电阻

Q11-9 (a) $A_1 X_{1,2} = A_2 X_{2,1}$ (互拉性)

求商系数 $\rightarrow$

(非12)

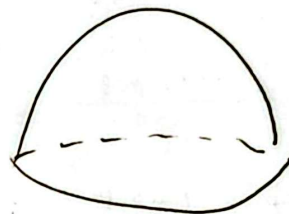
$X_{2,1} = 1$ ~~$X_{1,2} = 1$~~

$X_{1,2}$

~~$X_{1,2} = \frac{A_2}{A_1}$~~

若 $A_2 = 2\pi r^2$, $A_1 = \pi r^2$,

$X_{1,2} = 2$ #

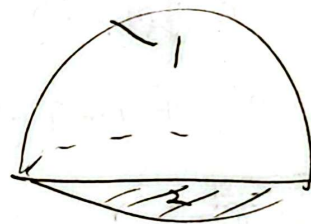


(b)

$A_1 X_{1,2} = A_2 X_{2,1}$ (互拉性)

(非12) $X_{2,1} = 1/4$

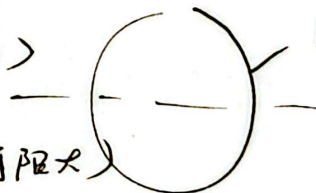
$\therefore X_{1,2} = \frac{A_2}{A_1} (X_{2,1}) = \frac{1}{2}$ #



(c)

$\rightarrow$ (仅 $\frac{1}{2}$ 球表面) 辐射到 (平面 z 上)

$X_{1,2} = \frac{1}{2}$, $X_{2,1} \approx 0$ ($\because A_2$ 无限大)

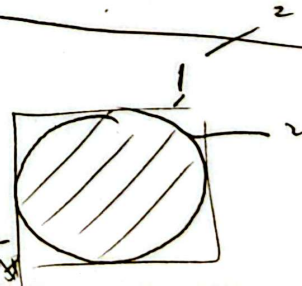


(d)

(A 切面) 2 球表面, $X_{2,1} = 1$

$A_1 X_{1,2} = A_2 X_{2,1}$

$X_{1,2} = \left(\frac{A_2}{A_1}\right) (X_{2,1}) = \frac{\pi r^2}{24\pi r^2} = \frac{\pi}{6}$ #



~~$6(2r)^2$~~

Q 11-13

传热学 2023050901

(无限大平板)

$$X_{1,2} = X_{2,1} = 1 \quad (\text{完全辐射表面})$$

$$q_{1,2} = \frac{E_{b1} - E_{b2}}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1}, \quad \sigma_b = 5.67 \times 10^{-8} \text{ W/m}^2 \text{K}^4$$

$$\begin{cases} E_{b1} = \sigma_b T_1^4 \\ E_{b2} = \sigma_b T_2^4 \end{cases}$$

$$\left(\frac{1-\epsilon_1}{\epsilon_1} + \frac{1-\epsilon_2}{\epsilon_2} + 1 \right)$$

$$\left(-1 + \frac{1}{\epsilon_1} \right) + \left(-1 + \frac{1}{\epsilon_2} \right) + 1$$

$$q_{1,2} = \frac{\sigma_b (T_1^4 - T_2^4)}{\frac{1}{\epsilon_1} + \frac{1}{\epsilon_2} - 1}$$

$$= \frac{(5.67 \times 10^{-8}) (73^4 - 323^4)}{\frac{1}{0.8} + \frac{1}{0.8} - 1}$$

$$\approx 734.03 \text{ W/m}^2 \quad (\text{Q. 11-12})$$

添加 $\epsilon = 0.1$ 透热板

$$R_{\text{surface}} \begin{cases} \frac{1-\epsilon_1}{\epsilon_1} = \frac{1-0.8}{0.8} = 0.25 \\ \frac{1-\epsilon_2}{\epsilon_2} = \frac{1-0.8}{0.8} = 0.25 \\ \frac{1-\epsilon}{\epsilon} = \frac{1-0.1}{0.1} = 9 \end{cases}$$

$$R_{\text{space}} = \frac{1}{X} \quad \therefore \text{not } AX$$

$$R_{\text{space}} = \frac{1}{X} = 1 \quad (\text{C. 11-12})$$

$$R_{\text{total}} = 0.25 + 1 + 9 + 9 + 1 + 0.25 = 21.5$$

$$q_{1,2} = \frac{\sigma_b (T_1^4 - T_2^4)}{R} = \frac{(5.67 \times 10^{-8}) (73^4 - 323^4)}{21.5} = 537.3 \text{ W/m}^2$$

Q 11-14

$$X_{12} = 0.2, X_{1,3} = X_{2,3} = 0.8$$

$$R_{s0} = \frac{1-\epsilon}{\epsilon A}$$

$$R_{s1} = R_{s2} = \frac{1-0.9}{0.9(1)} = 0.111$$

$$\frac{X}{L} = 0.1, \frac{Y}{L} = 0.1$$

$$F_{12} = X = 0.2, F_{11} = F_{11}$$

$$F_{12} + F_{13} = 1 \text{ (麦哲特)}$$

$$F_{21} + F_{23} = 1$$

$$\therefore F_{13} = F_{23} = 0.8$$

$$R_{s0} = \frac{1}{\epsilon A}$$

$$\therefore R_{s0} = \frac{1}{\epsilon A}$$

$$R_{12} = \frac{1}{1(0.2)} = 5$$

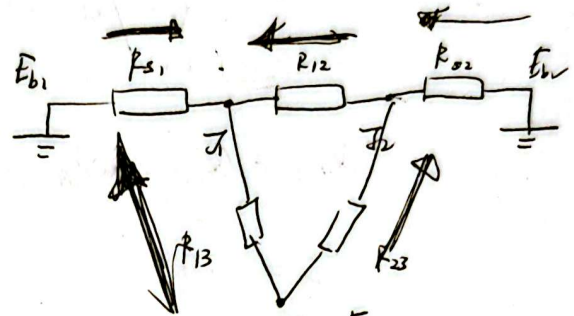
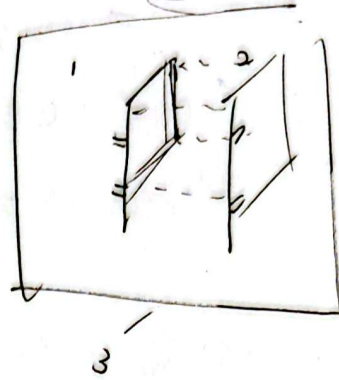
$$R_{13} = R_{23} = \frac{1}{0.8} = 1.25$$

$$E_{b0} = \sigma T_0^4, \sigma = 5.67 \times 10^{-8}$$

$$E_{b1} = 2.03 \times 10^4$$

$$E_{b2} = 2.85 \times 10^3$$

$$E_{b3} = T_3$$



若力里付

若力里付

$$R_{s0} = \frac{1-\epsilon}{\epsilon A}$$

$$\epsilon = 1$$



$$\begin{aligned} \text{①} \quad & \frac{E_{b1} - J_1}{R_{s1}} + \frac{J_1 - J_2}{R_{12}} + \frac{J_1 - J_3}{R_{13}} = 0 \\ \text{②} \quad & \frac{E_{b2} - J_2}{R_{s2}} + \frac{J_2 - J_1}{R_{12}} + \frac{J_2 - J_3}{R_{23}} = 0 \\ \text{③} \quad & J_3 = E_{b3} \end{aligned}$$

$$J_1 \approx 1.31 \times 10^4 \text{ W/m}^2$$

$$J_2 \approx 2.48 \times 10^3 \text{ W/m}^2$$

$$Q_1 = q_1 A = \left(\frac{E_{b1} - J_1}{R_{s1}} \right) A$$

$$Q_2 = q_2 A = \left(\frac{E_{b2} - J_2}{R_{s2}} \right) A$$

$$\begin{cases} Q_1 = -5.9 \text{ kW} \\ Q_2 = -0.37 \text{ kW} \end{cases}$$

Q4-17

$$Q_{14} = \alpha_s G_s$$

$$\begin{aligned} (Q_{14}) &= 0.5 (1000) \\ &= 500 \text{ W/m}^2 \end{aligned}$$

$$\begin{aligned} Q_{12} &= h (t - t_f) \\ (Q_{12}) &= 20 (t - 35) \end{aligned} \quad t_f = 35^\circ\text{C}$$

$$(Q_{12}) Q_{12} = 20 T^4$$

$$\begin{aligned} &= (5.67 \times 10^{-8}) (273 + t)^4 \\ (0.2) \end{aligned}$$

$$Q_{1,3} + Q_{1,2} = Q_{14}$$

$$t = 53.55^\circ\text{C}$$

③

対外
(放射)

